

ANVESHAK HUB PRIVATE LIMITED

Anveshak CRM

Product Blueprint & Low-Level Design (LLD)

A highly tailored customer relationship management application engineered for Anveshak Hub. This document integrates the initial business product specifications with a complete, production-grade low-level engineering specification. The system features a bifurcated architecture distinguishing between standard sales pipelines and a pro-tier automation, document engine, and audit governance layer.

PREPARED BY

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TARGET ENVIRONMENT

Next.js 15+ | React 19 | Neon Postgres | Clerk Auth | Prisma ORM

DATE OF COMPILATION

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PART I: Product Specification

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For: Anveshak Hub Private Limited - Internal Project (Month 1)

Stack: Next.js 16 + React 19 + Neon (Postgres) + Prisma + Clerk + shadcn/ui

1. Auth & User Management

1.1 Authentication

- Email/password + Google OAuth login via Clerk
- Invite-only signup (no public registration) - admin sends invite link
- Session persistence, auto-logout after inactivity (configurable, default 7 days)
- Forgot password flow (handled by Clerk)

1.2 Roles & Permissions

Role	Access
Admin	Full access - manage users, export data, view all deals/contacts, delete records
Manager	View team's deals/contacts, reassign leads, view reports, cannot manage users
Sales Rep	View/edit only their assigned contacts, leads, deals, tasks

1.3 User Profile

- Name, email, phone, role, profile photo
- "Last active" timestamp
- Team/department field (for future team-based filtering)

1.4 Audit Log

- Every create/edit/delete action logged: who, what, when, before/after value (for critical fields like deal stage, deal value)
- Admin-only view, filterable by user/date/action type

2. Contact Management

2.1 Contact Record Fields

- Full name, email, phone, alternate phone
- Company/organization name (linked to Company entity)
- Job title/designation
- Address (city, state - relevant for B2G/manufacturing clients across Karnataka)
- Source (Website, Referral, Hackathon, Cold Call, Event, etc.)
- Owner (assigned sales rep)
- Tags (custom labels - e.g., "Manufacturing", "Hot Lead", "Existing Client")
- Created date, last contacted date

2.2 Company/Account Entity

- Separate from individual contacts - one company can have multiple contacts
- Fields: company name, industry, website, size, address
- Linked deals and contacts roll up under the company profile

2.3 Contact List View

- Sortable table: name, company, owner, last contact date, score (see Lead Scoring)
- Filters: by owner, by source, by tag, by date added
- Search bar (name, email, phone, company - fuzzy match)
- Bulk actions: bulk tag, bulk reassign owner, bulk export

2.4 Contact Detail Page

- Header: name, company, contact info, edit button
- Tabs/sections:
 - **Activity Timeline** - chronological feed of all calls, emails, meetings, notes, stage changes
 - **Notes** - free-text notes, timestamped, author tagged
 - **Linked Deals** - all deals associated with this contact
 - **Tasks** - open/completed tasks tied to this contact

2.5 Duplicate Detection

- On contact creation, check for matching email/phone
- Show warning + option to "merge" or "create anyway"

3. Lead Management

3.1 Lead vs Contact (data model logic)

- A Lead is a pre-qualified prospect (not yet a confirmed contact/deal)
- On qualification, a lead converts into a Contact + Deal (one-click "Convert" action)

3.2 Lead Fields

- Name, company, email, phone
- Source (same as contact sources)
- Status: New ? Contacted ? Qualified ? Converted / Disqualified
- Owner
- Notes
- Score (see section 5)

3.3 Lead Capture

- Manual entry by sales rep
- Web form embed for inbound leads from swachbandhu.site-style landing pages

3.4 Lead List View

- Default sorted by Score (highest first) - surfaces hottest leads automatically
- Status filter tabs (New / Contacted / Qualified / Disqualified)
- Color-coded score badge: Cold (0-30) / Warm (31-60) / Hot (61+)

4. Sales Pipeline (Deals)

4.1 Deal Fields

- Deal name (auto-suggested: "{Company} - {Product/Service}")
- Linked contact + company
- Deal value (INR)
- Stage (customizable: New, Contacted, Proposal Sent, Negotiation, Won, Lost)
- Probability % (auto-suggested per stage, editable)
- Expected close date
- Owner
- Lost reason (dropdown, required when marked Lost - e.g., "Budget", "Competitor", "Timing")

4.2 Kanban Board View

- Columns = pipeline stages
- Drag-and-drop card between stages (auto-updates stage + logs activity)
- Card shows: company name, deal value, owner avatar, days in current stage
- Column header shows: total deal count + total INR value per stage

4.3 Deal Detail Page

- All deal fields editable inline
- Activity timeline (calls, emails, stage changes, notes - same pattern as Contact)
- Linked Quotes (see section 6)
- "Mark as Won/Lost" action with required reason on Lost

4.4 Pipeline Customization

- Admin can rename/reorder/add stages
- Custom stage colors

5. Lead Scoring (Rule-Based)

5.1 Scoring Rules (configurable table, not hardcoded in code)

Action	Points
Email opened	+10
Email link clicked	+15
Demo requested	+20
Call answered	+15
Call - no answer	-5
Meeting scheduled	+20
No activity in 7 days	-10
Marked "Not Interested"	-30 (auto-disqualify if score drops below 0)

5.2 Score Calculation

- Recalculated on every relevant activity event (not polled - event-driven)
- scoreBreakdown field stores a log of what contributed (for transparency - sales reps can see why a lead is scored a certain way)

5.3 Score Display

- Badge on lead list (Hot/Warm/Cold + numeric score)
- Score history mini-chart on lead detail page (optional, v2)

5.4 Admin Controls

- Admin can adjust point values per rule from a settings page (no code changes needed to tune scoring)

6. Document/Quote Generation

6.1 Quote Creation

- Generate from a Deal - pulls company name, contact, deal value automatically
- Line items: description, quantity, unit price, auto-calculated total
- Tax field (GST %, auto-calculated)
- Terms & conditions text block (reusable template)
- Validity date

6.2 Quote Statuses

- Draft ? Sent ? Accepted / Rejected / Expired

6.3 PDF Generation

- Branded template (Anveshak Hub logo, registered office details - reuse info from their letterhead)
- Generated via @react-pdf/renderer
- Download button + "Send via Email" button (attaches PDF, uses email integration)

6.4 Quote Tracking

- List of all quotes per deal, with status and timestamps
- Quote history visible to Manager/Admin for forecasting accuracy

7. Task Management

7.1 Task Fields

- Title, description
- Linked to: Contact / Lead / Deal (optional, can be standalone)
- Assignee
- Due date + time
- Priority (Low/Medium/High)
- Status (Open/Completed)

7.2 Task Views

- "My Tasks" - grouped into Overdue / Today / Upcoming
- Team Tasks view (Manager/Admin only) - filterable by assignee

7.3 Reminders

- In-app notification 1 hour before due time
- Overdue tasks highlighted in red, surfaced on dashboard

8. Activity Logging (Calls/Meetings)

8.1 Structured Call Log

- Linked to Contact/Deal
- Fields: Date/time, Duration (minutes), Type (Call/Meeting/Site Visit), Outcome (dropdown: Interested / Not Interested / Follow-up Needed / No Answer / Rescheduled), Next step (text), Next follow-up date

8.2 Why Structured (not free text)

- Enables reporting ("how many calls resulted in Follow-up Needed this week")
- Feeds into Lead Scoring automatically based on Outcome value

8.3 Activity Feed

- All activities (calls, emails, notes, stage changes, quote sent) appear in one unified, chronological timeline per Contact/Deal

9. Calendar

9.1 Views

- Month / Week toggle
- Color-coded by type: Task (blue), Call/Meeting (teal), Deal close date (amber)

9.2 Functionality

- Click date ? quick-create task/meeting
 - Click event ? opens linked Contact/Deal
 - (v2) Two-way sync with Google Calendar
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10. Reports & Dashboards

10.1 Dashboard (Home Page)

- Metric cards: Open Leads, Active Deals, Pipeline Value (INR), Tasks Due Today
- Mini pipeline funnel chart
- "My hot leads" widget (top 5 by score)

10.2 Reports Module

- Conversion Funnel - Lead ? Contacted ? Qualified ? Won, with drop-off % at each stage
- Sales Forecast - sum of (deal value x probability%) by expected close month
- Team Performance - deals won/lost per rep, average deal cycle time
- Activity Report - calls made, meetings held per rep per week

10.3 Export

- Every report exportable to CSV/PDF
 - Date range filter on all reports
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11. Data Export & Backup

11.1 Manual Export (Admin-only)

- Export Contacts, Leads, Deals, Activities as CSV - individually or as full bundle (ZIP)
- Export includes all custom fields

11.2 Scheduled Backup (v2, nice-to-have)

- Weekly automatic DB snapshot stored separately (Neon supports branching/snapshots natively)

11.3 Export Audit

- Every export action logged (who, when, what was exported) - important for data governance since this is a company asset

12. Notifications

12.1 In-App

- Bell icon with unread count
- Triggers: task due soon, deal stage changed by teammate, lead reassigned to you, quote accepted

12.2 Email (Pro-tier scope, optional Month 1)

- Daily digest of overdue tasks
- Weekly pipeline summary to Managers

13. Settings (Admin Panel)

- User management (invite, deactivate, change roles)
- Pipeline stage customization
- Lead scoring rule configuration
- Custom fields management (add custom field to Contact/Lead/Deal)
- Quote template editor (terms & conditions, branding)
- Data export & backup controls

14. Non-Functional Requirements

Area	Requirement
Performance	Page loads < 2s on standard broadband
Responsiveness	Fully usable on mobile browser (PWA installable)
Security	Role-based server-side route protection (not just UI hiding), HTTPS only
Offline	Basic offline view of cached contacts/tasks (PWA service worker)
Browser support	Chrome, Edge, Safari (latest 2 versions)

PART II: Low-Level Design

This document details the low-level architecture, module design, database schemas, and interface structures for Anveshak CRM. To support presentation and modular delivery, features are divided into **Standard** (Core Sales Tracking) and **Pro/Premium** (Enterprise Automation & Analytics) tiers.

1. System Architecture Overview

Anveshak CRM is built on a modern, serverless-first TypeScript stack designed for speed, security, and scalability.

- **Frontend & Routing:** Next.js 15+ (App Router) using React Server Components (RSC) for page loads under 2 seconds, paired with React 19 Server Actions for write operations.
- **Authentication & Session Management:** Clerk Auth with role-based routing guards.
- **Database & ORM:** Neon Serverless PostgreSQL with Prisma Client.
- **UI System:** Tailwind CSS + `shadcn/ui` components based on Radix UI primitives.
- **PDF Generation:** `@react-pdf/renderer` executing on serverless API routes.

2. Feature Tiering Map

2.1 Standard Tier (Core CRM)

Features required to track basic customer relationships and deals.

- **Authentication:** Clerk-managed signup (invite-only restriction checked via database lookup), login (Email/Password, Google OAuth), and user profiles.
- **Contact Management:** Core CRUD operations for Contacts and Companies, list view with search, sorting, and manual pagination.
- **Lead Tracking:** Simple Lead list with manual status changes (`New`, `Contacted`, `Qualified`, `Disqualified`) and a one-click button to convert a Lead to a Contact + Deal.
- **Sales Pipeline:** Kanban board representing Deal stages, supporting drag-and-drop movements and tracking Deal values (Rs.).
- **Task Management:** Personal task checklist ("My Tasks") with due dates, priorities, and status indicators.
- **Activity Logging:** Basic call/meeting notes logging with standard text outcomes.

2.2 Pro & Premium Tier (Sales Automation & Operations)

Advanced tools enabling team governance, document generation, and intelligence.

- **Dynamic Custom Fields:** Metadata engine allowing Admins to define custom fields for Contacts, Leads, and Deals via UI without running database migrations.
- **Rule-Based Lead Scoring:** Event-driven calculator scoring leads based on activity outcomes, preserving a granular `scoreBreakdown` audit.
- **Document & Quote Generation:** Automated line-item quote creation, PDF compilation (GST, terms template), and download/send suite.
- **Advanced Analytics & Reports:** Dynamic charts for Conversion Funnel, Pipeline Forecasting (deal value × probability), Team Performance, and CSV/PDF export with date range filters.
- **Audit & Governance Logs:** Complete audit logging capturing before/after JSON states of records, plus export logs for security audits.
- **Enterprise Administration:** User deactivation, role shifting, pipeline stage customization, and lead scoring rules management.

3. Database Schema Design (Prisma)

The PostgreSQL database uses JSONB columns to handle Pro-tier dynamic features like custom fields and lead scoring logs while maintaining strict relational schemas for standard features.

4. Key Module Implementations

4.1 Lead Scoring Engine (Pro/Premium Tier)

Scores are recalculated asynchronously inside Next.js Server Actions on events using a modular calculation service:

4.2 Dynamic Custom Fields Engine (Pro/Premium Tier)

Allows users to add fields dynamically without schema modification.

4.3 Automated Audit Logger (Pro/Premium Tier)

Runs inside database-modifying API routes/Server Actions to record states before and after modifications.

5. UI Layout & Component Organization

The folder structure segregates standard pages and components from pro-tier overrides.

6. Access Control & Authorization (Standard & Pro)

Role-based restrictions are applied server-side using Next.js Middleware and custom wrapper checks to ensure secure endpoints.

7. Performance & Quality Benchmarks

1. **Fast Initial Load (Standard):** Using Next.js static optimizations and code splitting, the initial dashboard and lead lists must load under 2 seconds on standard broadband connections.
2. **Referential Consistency:** Deletions of a Contact, Lead, or Deal automatically cascades to associated notes, tasks, and activities using foreign key definitions (`onDelete: Cascade`).
3. **Governance Auditing (Pro):** Every export operation registers an `ExportLog` directly associated with the executing user ID.